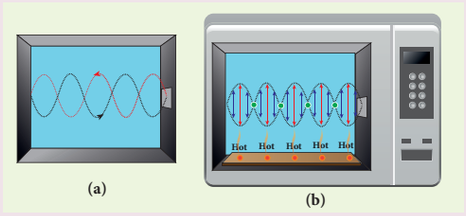
Activity

Measuring the speed of light using the microwave oven

Nowadays the microwave oven is very commonly used to heat the food items. Micro waves of wavelengths 1 mm to 30 cm are produced in these ovens. Such waves form the standing waves between the interior walls of the oven. It is interesting to note that the speed of light can be measured using micro wave oven.



(a) (b)



(c) (d)

We studied about the standing waves in XI physics, Volume 2, Unit 11. The standing waves have nodes and antinodes at fixed points. At node point, the amplitude of the wave is zero and at antinodes point, the amplitude is maximum. In other words, the maximal energy of microwaves is located at antinode points. When we keep some food items like chappathi or choclate (after removing the rotating platform) inside the oven, we can notice that at antinode locations, chappathi will be burnt more than other locations. It is shown in the Figure (c) and (d). The distance between two successive burnt spots will give the half wavelength of microwave. The frequency of microwave is printed in the panel of oven. By knowing wavelength and frequency of microwaves, using the formula $v\lambda = c$, we can calculate the speed of light c.

dehydrated fruits, in green houses to keep the plants warm, heat therapy for muscular pain or sprain, TV remote as a signal carrier, to look through haze fog or mist and used in night vision or infrared photography.

Visible light

It is produced by incandescent bodies and also it is radiated by excited atoms in gases. The frequency range is from 4×10^{14} Hz to 8×10^{14} Hz.

It obeys the laws of reflection and refraction. It undergoes interference, diffraction and can be polarised. It exhibits photo-electric effect also. It can be used to study the structure of molecules, arrangement of electrons in external shells of atoms. It causes sensation of vision.

Ultraviolet radiation

It is produced by Sun, arc and ionized gases. Its frequency range is from 8×10^{14} Hz to 10^{17} Hz.

It has less penetrating power. It can be absorbed by atmospheric ozone and is harmful to human body. It is used to destroy bacteria in sterilizing the surgical instruments, burglar alarm, to detect the invisible writing, finger prints and also in the study of atomic structure.

X-rays

It is produced when there is sudden stopping of high speed electrons at high-atomic number target, and also by electronic transitions among the innermost orbits of atoms. The frequency range of X-rays is from 10^{17} Hz to 10^{19} Hz.

X-rays have more penetrating power than ultraviolet radiation. X-rays are used extensively in studying structures of inner atomic electron shells and crystal structures. It is used in detecting fractures, diseased organs, formation of bones and stones,

12th Physics_Vol-1_Unit 5_English Medium.indd 276 24-12-2021 15:57:33